Fiscal Year:	FY 2016	**Task Last Updated:**	FY 08/21/2015
PI Name:	Wotring, Virginia Ph.D.		
Project Title:	Dose Tracker Application for Monitoring Crew Medication Usage, Symptoms, and Adverse Effects During Missions		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Virginia.Wotring@bcm.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	
Organization Name:	Baylor College of Medicine		
PI Address 1:	Center for Space Medicine		
PI Address 2:	6500 Main St, Suite 910		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77030	**Congressional District:**	9
Comments:	PI formerly with Universities Space Research Association until fall 2015.		
Project Type:	Flight	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/25/2013	**End Date:**	05/25/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date is now 5/25/2016 due to PI move to Baylor College of Medicine; continued by "Dose Tracker Application for Monitoring Crew Medication Usage, Symptoms, and Adverse Effects During Missions--NNX16AK78G" with the same investigator, Dr. Virginia Wotring.		
Key Personnel Changes/Previous PI:	August 2015 report: Kami Faust and Shahin Pour are no longer working on this project.		
COI Name (Institution):	Smith, LaRona M.S.,R.N. (JES Tech/NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Do medications used during spaceflights work the same as they do on Earth? This single question underlies most of the unknowns within NASA's Human Research Program Risk of Clinically Relevant Unpredicted Effects of Medication. During spaceflight, the body undergoes a number of physiological changes that are expected to result in altered interactions with administered medications, but it is not yet known if, or to what extent, these actually occur. The potential for therapeutically relevant alteration in either pharmacokinetics (how the body handles administered medications) or pharmacodynamics (how administered medications act upon the body) has long been a concern. This observational epidemiological study is a proactive step toward addressing this issue via regular direct questioning of crewmember volunteers, a model that the Johnson Space Center (JSC) Nutritional Biochemistry Discipline has proven to be both feasible and useful. A tablet- or handheld device-based questionnaire will be used to permit fast and efficient collection of data regarding crewmembers' medication use on a near real-time basis, eliminating the current problems associated with recall over periods of weeks. Specific questions regarding medication use (somewhat different from the questions that physicians ask regarding patient health) will be asked of each participating crewmember. The data collection process will be streamlined by using a flexibly programmed computerized survey application that leverages the limited medication choices aboard, the doses available, typical dosing frequency, and side effects associated with each medication to provide an individualized short questionnaire for each medication use by the crewmember. Coded (de-identified) data will be delivered weekly to a secure server on the ground for analysis by study investigators. Post-flight (after re-adaptation to Earth's gravity), each participating crewmember will repeat recording their medication usage, so that their ground medication usage frequencies, doses, and perceptions may be compared to those recorded during flight.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal
Research Impact/Earth Benefits:	
Task Progress:	From 10/25/2014 to 8/7/2015, the following was accomplished on Dose Tracker Application for Monitoring Crew Medication Usage, Symptoms, and Adverse Effects During Missions: Many software requirements were met during the past year. They include: 1) Software (ground and flight versions) were tested and finalized. 2) Changes derived from subject input from Human Exploration Research Analog (HERA) Campaign 1 Missions 2-4 were incorporated in the software. 3) The Software Requirements Document was finalized in December 2014. 4) Dose Tracker was presented at the Computer Resources Control Panel (CRCP) on 7/14/14. 5) Software Classification Assessment and Safety Criticality Determination (Class D) received Final Signature on 4/7/15. 6) Payload Display Review Team/Integrated Display and Graphics Standards (PDRT/IDAGS) group granted approval on 7/17/15. 7) JF 1704 – Software Classification and Safety Criticality was completed and signed. 8) Project Plan was complete and signed. 9) JPR 7150.2 Compliance Matrix was completed. We are working now on finishing up the Test Plan, Version Description Document (VDD), Certificate of Compliance (CoC), and the final testing before the official delivery to the SSC (Station Support Computers) group to be incorporated with the Espresso service pack for delivery to the International Space Station (ISS) by November 2015. Subject recruitment and data collection have begun. Since 10/25/2014, the PI (principal investigator) and/or co-Investigator have pitched the study to 8 crew members. Five crew members have consented; the other 3 have not yet been asked for final signatures, per usual International Space Station Medical Projects (ISSMP) practices. Ground data collection began in July 2015 (n=1). Ground data collection is planned to begin by 10/2015 for 2 other crewmembers. Two additional crewmembers have chosen to perform their ground data collection sessions after their flights. No flight data collection has occurred yet, since no flights have occurred for any of the crew approached for this study. Flight data collection is expected to begin in December 2015. Principal Investigator Dr. Virginia Wotring moved to Baylor College of Medicine (BCM) as of Jun 1 2015. Transfer of the study to BCM is in progress and is expected to be complete by October 2015. EDITOR's NOTE (7/28/2016): End date is now 5/25/2016 due to PI move to Baylor College of Medicine; continued by "Dose Tracker Application for Monitoring Crew Medication Usage, Symptoms, and Adverse Effects During Missions-NNX16AK78G" with the same investigator, Dr. Virginia Wotring.
Bibliography Type:	Description: (Last Updated: 12/24/2019)